



D Y PATIL
UNIVERSITY
DISCOVER • LEARN • GROW

TECHNOLOGY | TOURISM | SUSTAINABLE FUTURE

PRESENTS

AUGMENTED REALITY AND VIRTUAL REALITY IN TOURISM

TRANSFORMING TRAVEL EXPERIENCES



EXPLORE
VIRTUALLY



DISCOVER
BOLDLY



EXPERIENCE
BEYOND



A MORE
CONNECTED
WORLD

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SAME WORLD
NEW PERSPECTIVE

*"Explore
Beyond
Reality"*



HISTORY
CULTURE
NATURE
PEOPLE
EVERYWHERE

Taj Mahal
Agra, India
A symbol of love
and a UNESCO
World Heritage Site.



PEOPLE | PLACES | POSSIBILITIES

INTRODUCTION

WHAT IS AR & VR IN TOURISM?

TECHNOLOGY IS CHANGING THE WAY
PEOPLE TRAVEL AND EXPLORE
DESTINATIONS.

*Explore
Beyond
Reality*

*"Same Places
New Perspectives"*

TRAVEL
LEARN
EXPERIENCE
CONNECT



AUGMENTED REALITY (AR)

Augmented Reality (AR) adds digital information, images, sounds, or directions to the real world.



Navigation



Historical
Information



Translation



Discovering
Attractions



VIRTUAL REALITY (VR)

Virtual Reality (VR) allows users to experience places through a completely digital environment.



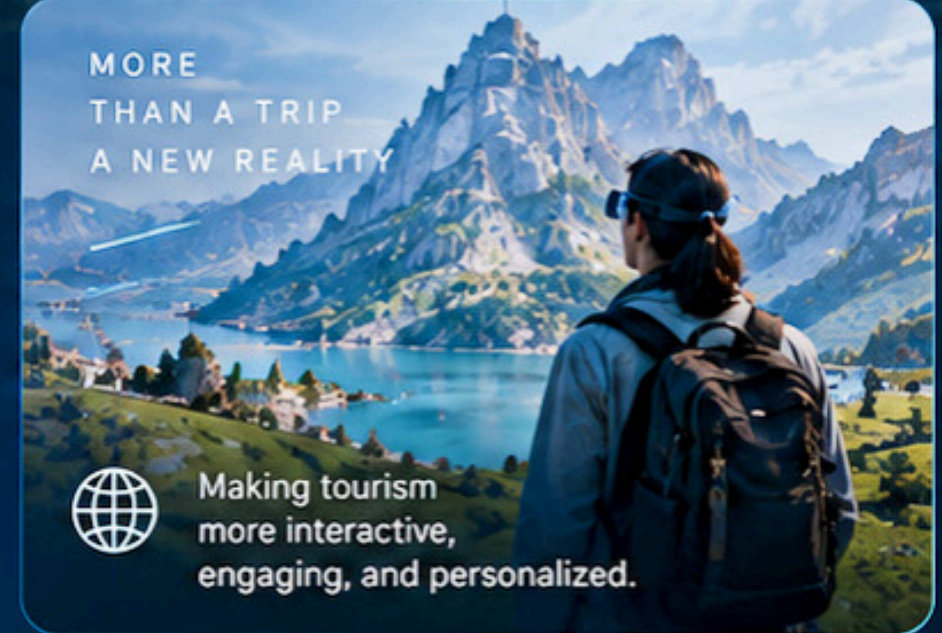
Explore
Virtually



Before
Travelling



Without
Physical Travel



MORE
THAN A TRIP
A NEW REALITY



Making tourism
more interactive,
engaging, and personalized.



Cultural Heritage

Preserve and experience
history like never before.



Accessibility

Makes destinations
available to everyone.



Education

Interactive learning
through real experiences.



Eco-Tourism

Supports conservation
and sustainable travel.

EVOLUTION OF AR & VR IN TOURISM

A JOURNEY FROM IDEAS TO IMMERSIVE EXPERIENCES

PAST
REAL PLACES

PRESENT
DIGITAL BRIDGES

FUTURE
LIMITLESS JOURNEYS

THE NEXT
DESTINATION
IS VIRTUAL

*"From
Imagination
to Exploration"*

TECHNOLOGY
CONNECTS
PEOPLE
PLACES
AND
CULTURES

1960s

Early Concepts



First head-mounted display (Sensorama, 1962) laid the foundation for immersive media.

THE IDEA BEGINS

1990s

Initial Development



Commercial VR devices like **Virtuality** systems emerged, mainly for gaming and simulation.

EARLY EXPERIMENTS

2000s

Maps & Guides



AR started appearing in navigation and travel guides, helping users explore locations.

DIGITAL ASSISTANCE

2010s

Mainstream Adoption



Smartphones and apps like Google Maps AR brought immersive experiences to travelers.

AR GOES MOBILE

2020s

Immersive Tourism

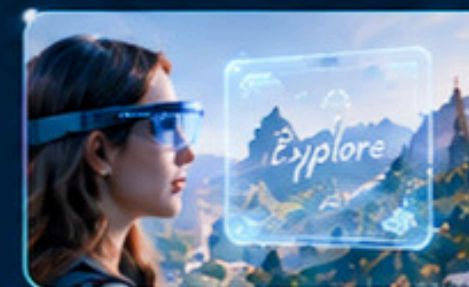


VR tours, 360° videos and AR experiences made destinations accessible worldwide.

TRAVEL GOES VIRTUAL

Today

Integrated Experiences



AR/VR integrated with AI, 5G, and spatial computing for real-time, personalized travel experiences.

A MORE CONNECTED WORLD

Future

Beyond Boundaries



Hyper-realistic virtual destinations, digital twin cities, and metaverse tourism will redefine how we explore the world.

LIMITLESS POSSIBILITIES

*"Same Curiosity.
A Different Reality."*



REAL PLACES
THEN



ENHANCED
EXPERIENCES
NOW



LIMITLESS
JOURNEYS
TOMORROW

DIFFERENT
TECHNOLOGY
SAME HUMAN DESIRE
TO EXPLORE.

TECHNOLOGIES USED IN AR & VR FOR TOURISM

REAL TECHNOLOGIES. REAL EXPERIENCES.

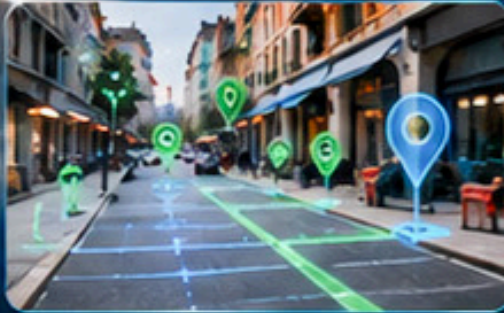
"Technology Brings Destinations Closer Than Ever"



SAME
PLACES
NEW
PERSPECTIVES



1. SLAM (Simultaneous Localization and Mapping)



Helps devices understand and map the surroundings in real-time for accurate AR experiences.



2. IMU (Inertial Measurement Unit)



Uses accelerometers and gyroscopes to track the movement, rotation and orientation of the device.



3. Cameras & Sensors



Capture real-world environments to overlay digital content in AR and create 3D experiences.



4. LiDAR (Light Detection and Ranging)



Measures distance using laser light to create detailed 3D maps of real-world spaces.



5. Photogrammetry



Creates 3D models from multiple photos, widely used to digitize heritage sites and landmarks.



6. NeRF (Neural Radiance Fields)



2D Photos → NeRF → 3D Virtual Model

Uses AI to create highly realistic 3D scenes from images, making virtual tours more immersive.



7. 3D Gaussian Splatting (3DGS)



A next-gen rendering technique that creates high-quality, real-time 3D environments from photos or videos.



8. 5G + MEC (Multi-access Edge Computing)



Provides ultra-fast, low-latency connectivity to deliver smooth and realistic AR/VR experiences, even on mobile devices.



9. AI & Cloud Computing



Enhances AR/VR tourism with intelligent recommendations, real-time translation and cloud-based virtual experiences.



IMPORTANCE OF AR & VR IN TOURISM

MORE THAN A TRIP, A BETTER EXPERIENCE

*"Explore
Engage
Preserve"*

*"Travel
Smarter
Experience
Deeper"*

PEOPLE
PLACES
CULTURES
CONNECTIONS



Better Trip Planning



Explore destinations virtually before traveling, helping tourists make informed decisions.



Enhanced Navigation



AR provides real-time directions, location-based information, and nearby attractions.



Preservation of Cultural Heritage



VR recreates lost or damaged heritage sites, keeping history alive for future generations.



Increased Accessibility



Allows people who cannot travel physically to explore places through immersive virtual experiences.



More Engaging Experiences



Transforms passive sightseeing into interactive and memorable experiences.



Boosts Tourism Industry



Creates new opportunities for businesses, attracts more tourists, and supports local economies.



CONNECTS
PEOPLE



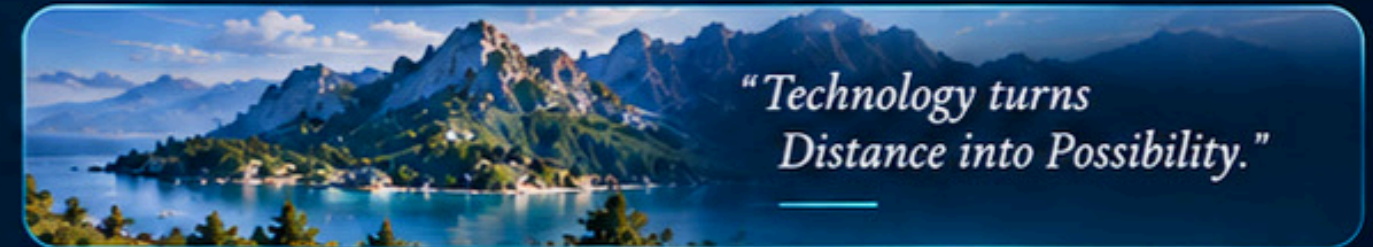
PRESERVES
HERITAGE



CREATES
OPPORTUNITIES



BUILDS A
BRIGHTER TOMORROW



*"Technology turns
Distance into Possibility."*

AR vs VR

A SIDE-BY-SIDE COMPARISON

TWO TECHNOLOGIES. ONE GOAL.
A MORE IMMERSIVE WORLD.

*"Different
Experiences
Same
Wonder"*

EXPLORE
COMPARE
UNDERSTAND
CHOOSE
EXPERIENCE



AUGMENTED REALITY (AR)

ADDS THE DIGITAL WORLD TO YOUR REAL WORLD

*Enhance
What's Real*

	What it is	Overlays digital information on the real world.
	User Experience	Sees and interacts with both real and virtual elements.
	Device	Smartphones, AR glasses, tablets.
	Environment	Real-world surroundings.
	Use in Tourism	Navigation, live translations, interactive guides, on-site information.
	Example	Point your camera at a monument to see its history, 3D model or reconstruction.



REAL PLACES. MORE STORIES.



VIRTUAL REALITY (VR)

TAKES YOU TO A COMPLETELY DIGITAL WORLD

*Step Into
Another Reality*

	What it is	Creates a fully immersive, computer-generated environment.
	User Experience	Completely immersed in a virtual world.
	Device	VR headsets (Meta Quest, HTC Vive, PlayStation VR, etc.).
	Environment	Virtual environment (no real-world view).
	Use in Tourism	Virtual tours, destination previews, heritage site reconstructions, immersive experiences.
	Example	Explore ancient Rome, walk through lost cities, or experience destinations before traveling.



ANY PLACE. ANY TIME.



Same Purpose

Make tourism more accessible, engaging and meaningful.



Complementary

AR enhances on-site travel, VR inspires and pre-explores.



Stronger Together

Combining AR & VR can create end-to-end immersive travel experiences.



A Brighter Tomorrow

More connected people, places and cultures.

APPLICATIONS OF AR & VR IN TOURISM

REAL USE CASES.. REAL IMPACT.

*"Technology
Takes You There
Before
You Travel"*

EXPLORE
WITHOUT LIMITS



SAME PLACES.
NEW PERSPECTIVES.



EXPLORE
EXPERIENCE
LEARN
CONNECT
PRESERVE



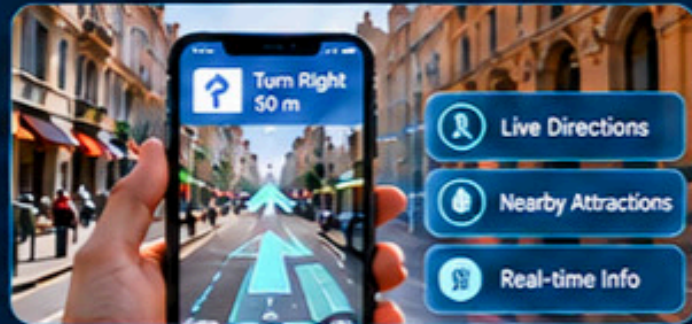
01 Virtual Destination Tours



Explore destinations in 360° before traveling, helping users plan better and make informed decisions.



02 AR Navigation & On-Site Assistance



Get real-time directions, location-based information, and useful suggestions while exploring new places.



03 Museums & Cultural Heritage



See history come alive with 3D reconstructions, interactive exhibits, and immersive storytelling.



04 Hotels & Accommodations



Take virtual tours of hotels and resorts, explore rooms, amenities, and surroundings before booking.



05 Events & Festivals



Experience cultural events, festivals, and live shows virtually, increasing global reach and participation.



06 Adventure & Nature Tourism



Virtually experience treks, wildlife safaris, and natural wonders, especially for hard-to-reach locations.



07 Educational Tourism



Interactive learning experiences for students, making educational trips more engaging and meaningful.



08 Marketing & Promotion



Tourism boards and businesses can use AR/VR to create immersive marketing campaigns.



09 Accessibility & Inclusive Travel



Helps people with disabilities or mobility issues experience destinations virtually and plan accessible travel.



10 Sustainable Tourism



Virtual experiences can reduce overtourism, promote responsible travel, and raise awareness about conservation.



EXPLORE MORE PLACES



ENGAGE MORE PEOPLE



PRESERVE OUR HERITAGE



CREATE A SUSTAINABLE FUTURE



*"More than Destinations,
Meaningful Experiences."*

BENEFITS & CHALLENGES OF AR & VR IN TOURISM

GREAT POTENTIAL. REAL CHALLENGES. A BRIGHTER FUTURE.

"Every Innovation Brings a New Perspective"

PEOPLE | IDEAS
PLACES | WITHOUT
CULTURES | LIMITS



KEY BENEFITS

CREATING RICHER, SMARTER AND MORE INCLUSIVE TRAVEL EXPERIENCES



Enhanced User Experience

Provides immersive and interactive experiences, making travel more engaging and memorable.



Accessibility

Lets people explore destinations virtually, helping those who cannot travel due to distance, health, or financial constraints.



Education & Awareness

Helps users learn about history, culture, and local traditions through interactive and realistic simulations.



Boosts Tourism Industry

Attracts more tourists, increases destination visibility, and creates new business opportunities.



Sustainable Tourism

Reduces unnecessary travel, promotes responsible tourism, and helps preserve cultural and natural heritage.



Personalized Experiences

Uses AI and data to provide tailored travel recommendations and interactive, location-based content.

BALANCING INNOVATION WITH REALITY



KEY CHALLENGES

OVERCOMING BARRIERS FOR WIDESPREAD ADOPTION



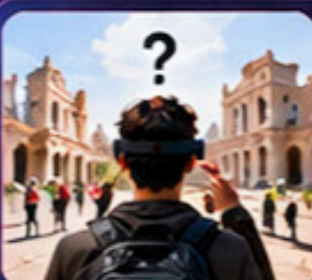
High Cost

Advanced hardware, software, and content development can be expensive for both providers and users.



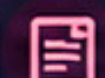
Technical Limitations

Issues like device weight, battery life, limited field of view, and need for high-speed internet can affect the experience.



Lack of Awareness and Adoption

Many potential users are still unaware of AR/VR benefits or hesitant to use new technology.



Limited Content Availability

High-quality, accurate, and engaging AR/VR content is still limited for many destinations.



Privacy and Data Security

Use of location data, personal information, and user behavior raises privacy and security concerns.



Reduced Real Travel Experience?

Some argue that virtual experiences might reduce the motivation to visit places in person.



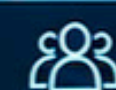
OPPORTUNITIES OUTWEIGH CHALLENGES WITH CONTINUOUS INNOVATION.



"Challenges Today, Better Journeys Tomorrow."



INNOVATE



COLLABORATE



PRESERVE



EXPLORE

FUTURE SCOPE OF AR & VR IN TOURISM

IMAGINE TODAY. EXPERIENCE TOMORROW.

*"The Next Destination
is a Smarter World"*



EXPLORE
BEYOND LIMITS

NEW
TECHNOLOGIES
NEW DESTINATIONS
A BRIGHTER
TOMORROW

01 AI-Powered Personalization



AI will create personalized travel recommendations using AR/VR based on user interests, past behavior, and real-time data.

02 Mixed Reality Travel Experiences



MR (Mixed Reality) will combine real and virtual worlds, allowing tourists to interact with destinations like never before.

03 Haptic & Multi-Sensory Technology



Future devices will include haptic gloves, spatial audio, and even smell simulation, making virtual travel feel more real.

- TOUCH
- SOUND
- SMELL
- TEMPERATURE

04 5G, 6G and Edge Computing



Ultra-fast connectivity will enable real-time, high-quality AR/VR experiences anywhere in the world without delays.

05 Digital Twins of Destinations



Complete digital replicas of cities and heritage sites will help in virtual exploration, preservation, and better tourism planning.

- EXPLORE
- ANALYZE
- PRESERVE
- PLAN

06 Space & Underwater Tourism



AR/VR will make extreme destinations like space, deep oceans, and inaccessible regions possible to explore virtually before real travel.

07 Cultural Preservation



AR/VR will help restore and recreate lost heritage sites, keeping culture and history alive for future generations.

PAST
PRESENT
FUTURE

08 Sustainable & Responsible Tourism



Virtual travel can reduce over-tourism, promote responsible travel, and create awareness about environmental conservation.

- AWARENESS
- LOWER CARBON FOOTPRINT
- VIRTUAL FIRST
- CONSERVE NATURE

09 Metaverse Travel Platforms



The metaverse will bring global destinations, people, and cultures together in shared virtual spaces for social and travel experiences.

TRAVEL
WITHOUT
BORDERS

10 Endless Possibilities



With continuous innovation, AR/VR will keep expanding the boundaries of tourism, creating experiences we can only imagine today.

*"The Journey
Never Ends."*



BOLDER
EXPLORATION



STRONGER
CONNECTIONS



A MORE
SUSTAINABLE WORLD



BRIGHTER
POSSIBILITIES



SAME PLANET.
NEW PERSPECTIVES.



TOWARDS A SMARTER, MORE CONNECTED TOURISM FUTURE

TECHNOLOGY. PEOPLE. PLACES. POSSIBILITIES.

*Same"
Places
New
Perspectives*



*"Tourism
Today.
A Brighter
Tomorrow."*

EXPLORE
LEARN
EXPERIENCE
PRESERVE
BELONG

01 TRANSFORMING TRAVEL EXPERIENCES



EXPLORE
BEFORE
YOU TRAVEL



Immersive pre-travel experiences



Better trip planning and decision-making



More engaging and memorable journeys



Connects people with diverse cultures

02 PRESERVING CULTURAL HERITAGE



PAST
PRESENT
FUTURE



Digitally preserves historical sites



Brings lost heritage back to life



Educates future generations



Promotes global awareness and respect

03 CREATING INCLUSIVE TOURISM



ACCESSIBLE
FOR EVERYONE



VIRTUAL TOURS



ASSISTIVE TECH



BARRIER-FREE INFO



Helps people with disabilities explore freely



Brings destinations to those who cannot travel



Promotes equal access to culture and nature



Makes tourism more humane and inclusive

04 DRIVING A SUSTAINABLE TOMORROW



PRESERVE
EXPLORE
RESPECT



Reduces over-tourism through virtual access



Supports responsible and eco-friendly travel



Creates new opportunities for local economies



Balances growth with environmental conservation



PEOPLE
CONNECTED



PLACES
EXPLORED



CULTURES
PRESERVED



IDEAS
REALIZED



A BRIGHTER
TOMORROW



*"More Than a Trip,
A Better Tomorrow."*

CONCLUSION & WAY FORWARD

IMMERSIVE TECHNOLOGY TODAY.
A MORE CONNECTED TOMORROW.

*"Technology
Bridges People,
Places and
Possibilities."*

REAL PLACES
BRIGHTER FUTURES

*Explore
Preserve
Connect
Grow*



KEY TAKEAWAYS

AR and VR have the power to transform tourism by making it more accessible, engaging, and sustainable.



Create immersive and memorable travel experiences



Increase accessibility for all travelers



Support local economies and global awareness



Help preserve cultural and natural heritage



MORE
THAN A VISIT
A CONNECTION



IMPACT ON STAKEHOLDERS

A WIN-WIN FOR A BRIGHTER TOMORROW.



Tourists

Richer, safer and more personalized experiences.



Local Communities

Economic growth and cultural recognition.



Environment

Reduced over-tourism and more sustainable travel.



Governments & Organizations

Better planning, promotion and global collaboration.



FUTURE POSSIBILITIES

THE JOURNEY HAS JUST BEGUN.



NEW
PLACES
NEW PEOPLE
NEW PERSPECTIVES



More realistic and affordable AR/VR devices



Integration with AI and personalized travel assistants



Real-time global virtual travel platforms



Greater focus on sustainability and responsible tourism



Stronger collaboration between technology, tourism and education



OUR VISION

A WORLD WHERE EVERYONE CAN EXPLORE, EXPERIENCE AND APPRECIATE THE BEAUTY OF OUR PLANET — ANYTIME, ANYWHERE.



EXPLORE



LEARN



EXPERIENCE



PRESERVE



BELONG



PEOPLE
STRONGER TOGETHER



PLACES
MORE CONNECTED



POSSIBILITIES
BEYOND LIMITS



A SUSTAINABLE
TOMORROW

*"Different Experiences
Same Wonder
A Brighter Tomorrow."*